

with auto dialing facility to run the anti-theft system. The auto dialer of the mobile is triggered automatically by a relay and sensor mechanism. On sensing a theft threat, an alert call is given to a prestored number automatically. We are trying to tie up with mobile phone companies to get mobile handset with such modifications," says Ostwal.

Long life. A long life span makes Nano Ganesh a worthwhile investment for the pennywise farmer. "It is life-tested in the hazardous areas using specially developed simulation standards/ techniques by Ossian Agro Automation. The predicted life-span of the gadget is five years," says Ostwal.

Sowing seeds of Nano Ganesh

But what sweat went behind this innovation? It took almost four years to establish and convert Nano Ganesh into a techno-commercial successful unit, recalls Ostwal, who attributes the success to the four-member strong R&D team. "The R&D for Nano Ganesh is

based on the closed-loop system. Ideas in mind would first be tried in the lab and then tested on the field for some days with a local farmer as a user for the feedback. On the field, we had to wait for a long time for the power supply to come, many times spending the whole night in dense bushes near the water zones," says Ostwal.

Further, the path wasn't easy. Field challenges were hard nuts to crack. "It was difficult to get the system functioning in wet areas. There are lizards and rats in the panels. Further, 440V 3-phase wiring is very open and susceptible to shock hazards. Sparks in the starters of higher-HP motors could create a nuisance to automation systems. Voltage range is also extreme, i.e., single-phase 120V to 250V, or 3-phase 250V to 500V," says Ostwal.

Ossian Agro Automation had to take great care in designing the basic transformer to withstand high stresses at high voltage conditions. "Transient protectors like MOV and RFI filters were used in the primary side of the

transformer. The capacitors we used could withstand high voltages. We also chose ICs that had tolerance levels of 5-12 volts and introduced 0.1µF disk capacitors at places in the circuits. Overall, the circuit was designed in such a way that the load current extracted was the lowest, reducing the heat generated by the system," says Ostwal.

In the future!

Looking ahead, Ostwal believes Nano Ganesh technology could be used in controlling any electrical appliance. "Street lights and neon signboards, fountains and waterfalls, air-conditioners, industrial heaters or security systems—Nano Ganesh technology can be the remote to control them all," says Ostwal. Thus it is only a matter of time before mobiles become remote controls for life management. A mobile unlocking your door or opening your garage might just be on its way! ●

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